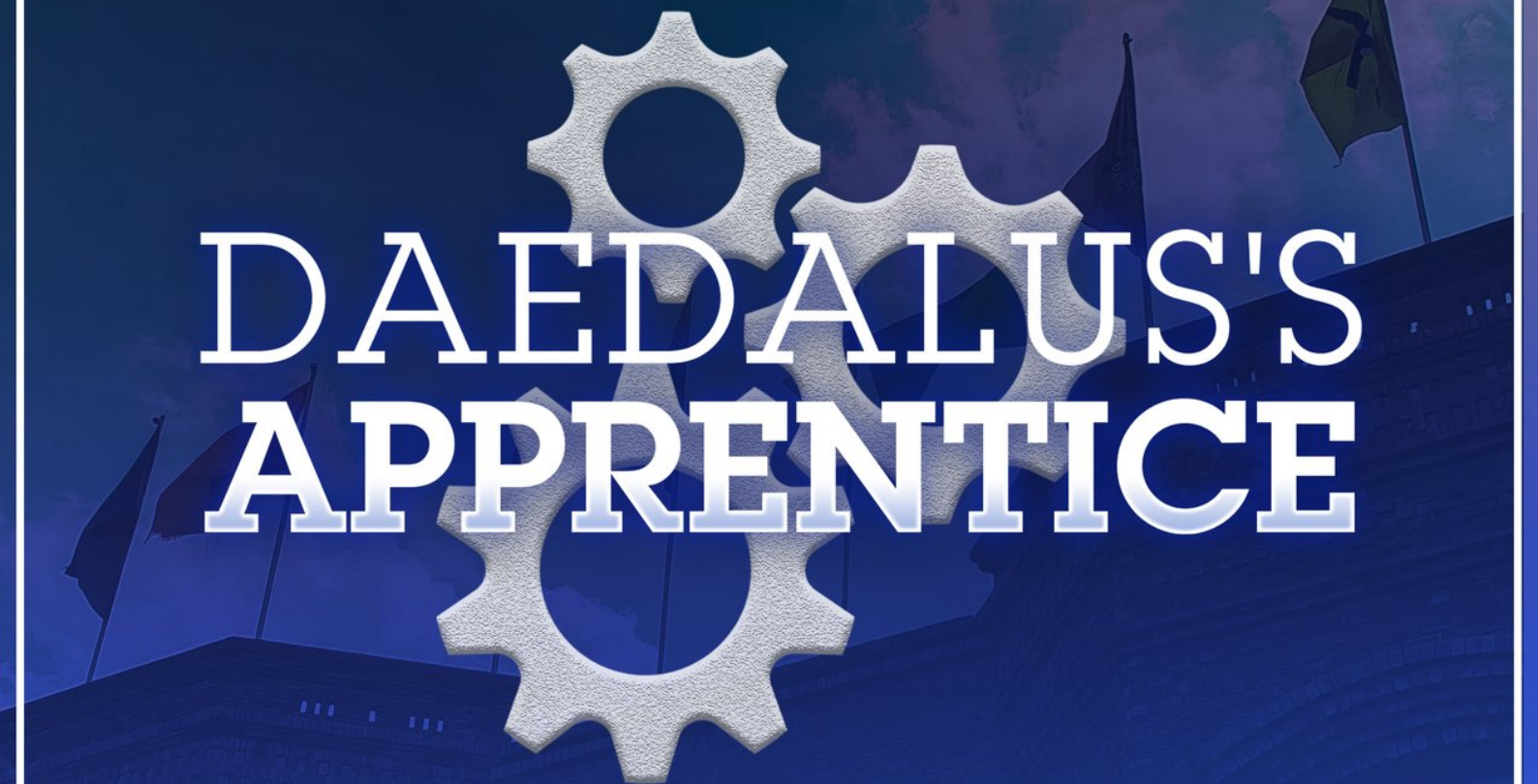
Lahore Grammar School
Johar Town Senior Boys Campus

DAEDALUS'S APPRENTICE

Study Guide



11 . 12 . 13
SEPTEMBER

LAHORE GRAMMAR SCHOOL JOHAR TOWN FOR BOYS
INNOVENTIONS '26
INDULGE · INVENT · INNOVATE

Round 1
Preliminary Presentation Round (Non-Elimination Round)

Delegates per team: No limit
Each team will be required to deliver a short presentation focusing on sustainable engineering solutions that can be applied in everyday life. The presentation should highlight practical, innovative, and feasible ideas that promote sustainability in areas such as energy usage, waste management, transportation, water conservation, and daily lifestyle practices.

Teams are expected to demonstrate clarity of thought, creativity, and basic engineering understanding while proposing solutions that are realistic and impactful.

Evaluation Criteria
a) Understanding of sustainability concepts – 20 marks
b) Innovation and creativity of proposed solutions – 20 marks
c) Practical feasibility of ideas – 30 marks
d) Clarity and confidence of presentation – 10 marks
e) Logical reasoning and problem-solving approach – 20 marks

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Round 2 and 3
Surprise Engineering Challenge (Non-Elimination Round)

Delegates per team: No limit

Round 2 will be a **Surprise Engineering Challenge** designed to test delegates' ability to apply fundamental engineering principles to an unfamiliar problem.
At the beginning of the round, each team will be presented with a **construction challenge** requiring them to design and construct a functional structure using a limited set of materials provided by the organizers.

The exact structure to be constructed will **not be revealed beforehand**. Depending on the challenge assigned, delegates may be required to construct structures such as a **bridge, building, tower, crane, shelter, platform, or other engineered structure**.

Materials Provided:

- Cardboard sheets
- Bamboo skewers
- Cutter
- Tape/glue (including hot glue gun with glue sticks)
- Rubber bands
- Paper sheets
- Scissors
- Pencil
- Ruler

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Addition to the Round:

At the beginning, every team receives an equal material kit. Teams can purchase extra materials from the "Engineering Market" if they want to improve their design.

However,

- Every additional purchase results in a point deduction.
- Teams must balance resource management with innovation.

Example (tentative);

- Extra cardboard: -5 points
- Extra skewers: -3 points
- Extra tape: -2 points

This tests real-world engineering decisions where resources are limited.

In Round 3, teams will have more time to perfect their builds and testing will take place subsequently with judges awarding marks to each team based on the judging criteria. The engineering constraints and judging criteria will be revealed to the delegates on time.